Title: **Effect of aerobic exercise on cancer-associated cognitive impairment: A proof-of-concept RCT**

Running head: Effect of exercise on cancer-related cognitive changes

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Abstract:

Background: Change in cognitive ability is a commonly reported side effect by breast cancer survivors (BCS). The underlying etiology of cognitive complaints is unclear and to date there is limited evidence for effective intervention strategies. Exercise has been shown to improve cognitive function in older adults and animal models treated with chemotherapy. This proof-of-concept randomized controlled trial (RCT) tested the effect of aerobic exercise versus usual lifestyle on cognitive function in postmenopausal BCS.

Methods: Women, age 40-65 years, postmenopausal, stage I-IIIA breast cancer, and who self-reported cognitive dysfunction following chemotherapy treatment were recruited and randomized to a 24-week aerobic exercise intervention (EX; n=10) or usual lifestyle control (CON; n=9). Participants completed self-report measures of the impact of cognitive issues on quality of life (FACT-Cog), objective neuropsychological testing and functional magnetic resonance imaging (fMRI) at baseline and 24-weeks.

Results: Compared to CON, EX had a reduced time to complete a processing speed test (Trail Making Task-A) (-14.2 seconds, p<0.01; effect size [ES] 0.35). Compared to CON, there was no improvement in self-reported cognitive function and effect sizes were small Interestingly, lack of between-group differences in Stroop behavioral performance were accompanied by functional changes in several brain regions of interest in EX compared to CON at 24-weeks.

Conclusion: These findings provide preliminary proof-of-concept results for the potential of aerobic exercise to improve cancer-related cognitive impairment, and will serve to inform the development of future trials.

Keywords: exercise, cognitive function, breast cancer, intervention, survivorship

Change in cognitive ability is reported by 15-75% of breast cancer survivors (BCS) who receive chemotherapy[1]. While these changes are most commonly concurrent with chemotherapy treatment and start to resolve within 6 months following treatment, for up to 35% of women these changes persist[2, 3]. In recent meta-analyses, BCS treated with chemotherapy perform more poorly than BCS who have not received chemotherapy and non-cancer controls, on neuropsychological tests of working memory[4], processing speed[4], motor performance[4, 5], spatial ability[4-6], and verbal ability[4-6]. These changes have a negative impact on quality of life, social roles and return to work[7]. The underlying etiology of cognitive complaints is unclear[8], but emerging data points to structural and functional changes in the brain as a possible etiology[8, 9].

Effective intervention strategies to address cognitive changes remain elusive. Mixed results have been reported for cognitive rehabilitation approaches (i.e., programs aimed at improving cognitive abilities, functional capacity, real-world skills, and/or internal metacognitive strategies), pharmaceutical agents (i.e., agents used to treat Alzheimer's disease or cancer-related fatigue), natural health products (e.g., ginkgo biloba) or mind-body therapies (i.e., tai chi and QiGong)[10, 11]. As a result patients typically receive little or no clinical advice on how to manage these symptoms.

There is convincing evidence that exercise reduces cognitive decline in healthy older adults[12] and those with mild cognitive impairment[13]. Of relevance, the cognitive functions that benefit from exercises[14] are comparable to the cognitive domains in which BCS are impacted, including executive function[4], attention[15], and processing speed[4]. Emerging research also demonstrates that aerobic exercise can alter brain structure and function[16]. These findings have sparked an interest in exercise as a potential intervention for cancer-related cognitive impairment, and suggest that brain imaging may be an effective tool to measure these subtle cognitive deficits.

In rodents treated with 5-fluorouracil and oxaliplatin, Fardell et al. reported[17] poorer performance on cognitive tasks in treated versus untreated control rodents. In contrast, treated rodents randomized to voluntary exercise (ran approximately 5 kilometers/night) displayed preserved cognitive function, particularly novel object recognition and spatial reference memory. These findings are consistent with those of Winocur et al.[18] who tested the effect of exercise, using voluntary wheel running following administration of 5-fluorouracil and methotrexate in rodents. Exercise (approximately 4 kilometers/day) prevented the significant decrease in performance on cognitive tasks and a reduction in hippocampal neurogenesis seen with chemotherapy administration in the non-exercise group. In humans, a recent observational study in early stage BCS treated with chemotherapy found a moderate association ($r=0.47$, $p<0.01$) between self-reported physical activity and visual memory abilities, but no association was seen with aerobic fitness[19].

To date no study has reported on the effect of an aerobic exercise intervention on self-reported cognitive complaints or objective neuropsychological tests in cancer survivors. We undertook a proof-of-concept RCT testing the effect of a 24-week aerobic exercise intervention compared to usual lifestyle control on measures of cancer-associated cognitive impairment in early stage BCS reporting persistent cognitive concerns. We hypothesized an improvement in self-reported measures of cancer-associated cognitive impairment, along with potential improvements in objective neuropsychological tests of verbal memory, working memory, processing speed, verbal fluency and executive function in the intervention group compared to control. However, given the reported inconsistencies and lack of agreement between self-report and objective measures in the literature, functional brain imaging was included as an outcome measure as an exploratory approach to capture changes that may not be captured by the other outcome measures.

Methods

Participants and recruitment:

Women, age 40-65 years, who self-report cognitive impairment following chemotherapy treatment for stage I-IIIA breast cancer were recruited through oncologist referral, posters and word of mouth (January 2011- June 2013). The recruitment goal was 30 women. All participants were ≥ 3 months and up to 3 years post adjuvant treatment, physically able to undertake an exercise program, postmenopausal (natural or chemotherapy-induced) at time of enrolment and receiving anti-hormonal treatment (i.e., aromatase inhibitor). As there is no available recognized short screening tool for cancer-associated cognitive impairment currently available, to screen for self-reported cancer-associated cognitive impairment, participants had to endorse at least one of six items on a list of common complaints developed by the research team (Supplement 1). Exclusion criteria were: > 90 min/week of self-reported moderate-vigorous physical activity (last 6 months); mini-mental status examination < 23; co-morbid conditions that could alter cognitive testing results, such as a psychiatric conditions, history of substance use disorder, or other neurological disorder (i.e., head injury, epilepsy, neurodegenerative disease); deemed unsafe for magnetic resonance imaging (MRI). Experimental procedures were approved by the University of British Columbia (UBC) Ethics Board and all participants provided written informed consent.

Study protocol:

Participants completed two study visits and questionnaires at baseline and end-of-study (24-weeks). Visit one consisted of a neuropsychological test battery, administered by trained study staff, blinded to group assignment, followed by anthropometrics (height and weight) and a graded maximal treadmill exercise test to measure exercise capacity. Visit two was an optional MRI session. Following completion of baseline measures, eligible participants were randomized using permutated blocks of four to six in a 1:1 ratio to the aerobic exercise intervention group (EX) or delayed exercise control (CON).

The primary outcome was self-reported cognitive dysfunction and associated impact on quality of life, using the Functional Assessment of Cancer Therapy-Cognitive Version 3 (FACT-Cog)[20]. The secondary outcomes were scores from a standardized neuropsychological test battery based on the recommendations of the International Cognition and Cancer Task Force (ICCTF)[21]. This included the Hopkins Verbal Learning Test-Revised for verbal memory and learning, Trail Making Test for processing speed and mental flexibility, "FAS" and animal naming for verbal fluency. The Stroop test was performed in the MRI scanner. It was selected as it been shown to be sensitive to capture change in cognitive function in exercise interventions in older adults[22] and those with mild cognitive impairment[23]. This test measures prepotent response inhibition, which is an executive function involved in ignoring irrelevant information and adaptively responding to current goals. Response inhibition is positively linked to performance-based measures of daily life functioning[24], a deficit noted among BCS[25]. As fatigue, depression and anxiety are all specific cancer-related symptoms that may impact cognition, these were assessed using FACT-fatigue (FACT-F), the Center for Epidemiology Studies Short Depression (CES-D 10) Scale and the Spielberger State Anxiety Scale, respectively.

Stroop Test, fMRI Data Acquisition and Preprocessing

The Stroop test requires participants to respond to the ink color of the word via three separate button presses corresponding to the three colors (red, blue, or green) as quickly and accurately as possible. This fMRI event-related design, consisted of 180 trials with equal numbers of stimuli (i.e., 60) in each of three conditions: incongruent (i.e., color-word printed in a color not denoted by the name), congruent (i.e., name matched ink color), and neutral (i.e., non-color words printed in one of the three colors); along with 60 null trials (i.e., only a crosshair (+) presented). To assess prepotent response inhibition, we compared response times (RTs) between neutral and incongruent trials[26], with faster response times in neutral relative to incongruent trials indicating the Stroop semantic interference effect; the meaning of the word is the cause of the interference. Difference in RT is the primary measure in a Stroop test[26]; a smaller time difference indicates better response inhibition, which has been linked to cognitive functioning in daily life and therefore improved cognitive functioning[24]. Mean accuracy rate was also computed.

Stimuli were presented using EPrime software (Psychology Software Tools, Pittsburgh, PA) on a MRI compatible back projection system (Panasonic LCD projector, model PT-L75OU). Words were presented for 2000 ms, followed by a 1000 ms interval crosshair presentation. The duration of null trials was jittered between 1500 and 7500 ms[27]. Each trial was first-order counter-balanced across the entire session.

Participants were scanned at the UBC MRI Research Centre on a Philips Achieva 3.0 T whole body MRI scanner (Phillips Healthcare, Andover, MD) using a sensitivity encoding head coil (SENSE). Prior to functional imaging, high-resolution T1-weighted axial images were acquired for anatomic localization and co-registration (TR = 7.7 ms, TE = 3.6 ms, bandwidth = 2.269 pixels (or 191.5 Hz per pixel), voxel size = 1 × 1 × 1 mm). Functional imaging data were collected as echo-planar images parallel to the AC-PC line using a gradient-echo pulse sequence, with repetition time (TR) = 2000 ms, echo time (TE) = 30 ms, flip angle = 90°, field of view (FOV) = 240 mm, matrix 128 mm × 128 mm. Each volume consisted of 36 axial slices of 3 mm thickness and a gap thickness of 1 mm. A total of 374 volumes were acquired during the Stroop test.

Functional MRI data processing was performed using Analysis of Functional NeuroImages (AFNI) software (GNU, General Public License)[28]. All functional data were spatially registered in three-dimensions to correct for head motion artifacts. Subjects with head motion exceeding 2 mm in any direction were excluded. A deconvolution analysis was used to construct a random-effects General Linear Model (GLM) in order to produce impulse response functions (IRFs) of the BOLD signal on a voxel-wise basis. Anatomical and functional images were co-registered and transformed to Talairach stereotaxic coordinate space, then spatially smoothed with a 4 mm full-width-half-maximum Gaussian kernel. Functional data were next converted to percent signal change (PSC) by repetition time. PSC was estimated on a voxel-wise level across the whole brain for each participant. The Talairach atlas[29] and MRI Atlas of the Human Cerebellum[30] were referenced to identify activation within the cerebral cortex and cerebellum, respectively.

Intervention

The 24-week intervention consisted of 150 min/week of moderate-to-vigorous aerobic exercise with two 45-minute supervised sessions per week in a research gym, and two additional 30 minute unsupervised home sessions (i.e. walking or activities of the participant's choice). Exercise intensity was based on each participants' baseline exercise test results, and used heart rate reserve (HRR) along with the ACSM metabolic equation for treadmill walking[31]. The prescription began at 60% of HRR for 20 minutes, with a weekly increase in duration toward 45 minutes by week six, followed by a weekly increase in intensity toward 80% HRR by week 12. Overall rating of perceived exertion (RPE) and average heart rate (Polar Canada, Lachine QC) was recorded for both supervised and home-based exercise sessions. A graded exercise test completed at week 12 was used to adjust the exercise intensity prescription overall and for interval training. Intervals consisted of four sets at a higher intensity (5 min-weeks 13-18 and 10 min-weeks 19-24) and five minutes at an easier intensity. Participants randomized to CON were asked to maintain usual lifestyle and offered a 12-week exercise program upon study completion.

Data analysis:

Independent t-tests and chi-square tests were used to test for differences in baseline characteristics for continuous and categorical variables, respectively. ANCOVA, with adjustment for baseline values only, was used to examine differences in primary and secondary outcomes at end of study. Effect sizes are presented as partial eta-squared. Statistical analyses were conducted using SPSS Statistics for Windows, Version 22.0. (IBM Corp., Armonk, NY). For fMRI data, we selected a priori regions of interest based on the results on our baseline fMRI data analyses that included all patients[31], as well as the neuroimaging literature employing the Stroop test[32]. Specifically, we identified all the regions from the baseline analyses that were relevant to the Stroop test, which included the right medial frontal gyrus (rMFG), right anterior cingulate cortex (rACC) and left superior frontal gyrus (lSFG). In line with our behavioral analyses, we computed the difference between incongruent and neutral trials of the Stroop test for every voxel in the brain. An ANOVA on the PSC of the contrast at the second TR (i.e., 2000 ms) following stimulus onset was used to examine the three regions associated with group differences over time in brain activation[33]. The omnibus ANOVA included a between-subject factor of Group (EX vs CON), and a within-subject factor of Time (baseline vs. end-of-study). To minimize family-wise error and control for multiple comparisons, the threshold for statistical significance was set at a p-value of 0.01 (critical F = 13.74) and a minimum cluster size of 200 contiguous voxels was employed[34]. PSC of each of the three regions of interest was extracted and submitted to a MANCOVA, which included a between-subject factor of Group (EX vs CON), end-of-study

PSCs of all three regions of interest as dependent variables, and controlled for baseline PSCs in those same regions as covariates.

The relationship between neural activity and behavioral performance in the Stroop test was examined as separate Pearson bivariate correlations between the PSC of the incongruent-neutral contrast of significant regions of interest and the incongruent-neutral RT difference at end-of-study. The neural and RT measures at baseline were partialled out in the correlational analyses.

Results:

Clinical and demographic characteristics:

Of the women screened, 19 were enrolled and randomized (EX, n=10 and CON, n=9) (Figure 1). Participants were on average of 52.4 years old, overweight, well-educated, working full-time and had low physical fitness (Table 1). The majority were within 12 months of completing adjuvant treatment. Three participants declined to participate in the fMRI portion, and two participants' fMRI data were excluded due to excessive head motion. Therefore, fMRI analyses were performed on the remaining 14 participants (EX, n=7 and CON, n=7).

Exercise Adherence in the Intervention Group

Participants attended 88% of supervised gym sessions (mean 1.8 sessions/week and 87.5 minutes/week) and participants met 82% of the prescribed exercise targets (mean intensity 74.5% HRR). Home session completion was 87% (mean 2.4 sessions/week and 101.5 minutes/week) and participants met 87% of the prescribed exercise targets (mean intensity 73.5% HRR). Compared to CON, aerobic fitness was significantly higher at 24-weeks in EX (adjusted mean difference +3.6 ml/kg/min, $p<0.01$; ES=0.43) (Table 2). There was no change in body weight. At 24 weeks, there was a clinically meaningful important difference in fatigue between EX and CON (FACT-F, 3-4 points)[35] that did not reach statistical significance (adjusted mean difference +6.3, p=0.18; ES=0.11). No adverse events occurred.

Change in self-report and objective neuropsychology tests

Compared to CON, there was no statically significant improvement in FACT-COG for EX at 24-weeks and effect sizes were small (Table 2). For objective neuropsychological tests, EX had faster completion for TMT-A (-14.2 seconds, p<0.01; ES= 0.35) and TMT-B (-18.6 seconds, p=0.29; ES=0.07), but overall no changes were noted with objective neuropsychological tests. There was no significant difference in the Stroop interference behavioral effect between groups at baseline (EX mean: 115ms, CON mean: 98ms; p=0.66), end of study (EX mean: 99ms, CON mean: 108ms; p=0.86) or change from baseline to end of study (p=0.52). The mean overall accuracy for all participants was very high at both baseline (96.3%, sd=2.6%) and end of study (97.7%, sd=1.8%). The change in overall task accuracy over time did not significantly differ between groups (p=0.84). For the fMRI data, all three regions of interest (i.e. rMFG, rACC and lSFG) showed significant Group-by-Time interaction (F=13.74, p=0.01) in the Stroop incongruent-neutral contrast. Although the main effect of Group was not statistically significant (Wilks' Δ= 0.96, $F(3,7)$=0.11, p>0.50), EX showed less PSC in the hemodynamic response in right cingulate cortex and left superior frontal gyrus, while CON showed less PSC in right medial frontal gyrus. Controlling for baseline group differences, the correlational analyses indicated that none of the PSC at any of the regions of interest correlated with the RT measure at end-of-study (p's > 0.40). These results suggest that some brain regions were less active in the EX relative to the CON group after the intervention, while accounting for group differences at baseline.

Discussion

This is the first RCT testing the effect of an aerobic exercise intervention to improve cognitive function in BCS reporting persistent cognitive changes following treatment. This proof-of-concept pilot study provides preliminary evidence for the potential of aerobic exercise to improve cognitive symptoms impacting quality of life and objective neuropsychological test performance in the areas of verbal fluency and motor processing speed. However, with the exception of TMT-A, these changes did not reach statistical significance and had small effect sizes.

Although the main group effect in the neuroimaging data did not reach significance, there was a notable pattern of neural activation between the two groups. At 24 weeks EX showed less brain activation over two regions implicated in high-level cognitive functions. These regions are the cingulate cortex and superior frontal gyrus, both of which are implicated in conflict monitoring – a key aspect of the Stroop test[36]. This finding is consistent with a large intervention study reporting that exercise decreased activation of the cingulate cortex in healthy older adults[16]. The reduced neural activity found in this region in EX accompanied by similar behavioral performance potentially suggests that following the intervention less mental effort was needed to maintain the same level of cognitive task performance, indicating greater cognitive efficiency to compensate for underlying cognitive deficits[37]. Given the small sample size in this study, these findings warrant replication in future studies.

A goal of this proof-of-concept study was to inform future trials, and identify outcome measures that may be most sensitive to change. There are several potential reasons for the small effect sizes seen in this study. Despite excellent attendance, adherence and improved aerobic fitness, the aerobic exercise intervention itself may not have been sufficient to induce change. Similar improvements in fitness has resulted in cognitive change in older adults[38] and future trials should look to this literature to inform the exercise prescription used as the exercise prescription required to counteract chemotherapy-associated cognitive changes in humans is unknown. In selecting the outcome measures, the study team chose to include the key tests, recommended by the ICCTF[18] and a well-established executive function task for the fMRI analysis (i.e., Stroop test). However, it is unclear if the test battery used is sufficiently sensitive to capture improvements in women with self-reported cognitive complaints following breast cancer. Finally, the ability to observe statistically significant change is likely limited by the small sample size, which is sensitive to any heterogeneity in the responses of participants. The lack of established diagnostic criteria makes it difficult to determine the most appropriate eligibility criteria for cognitive concerns. However, EX and CON participants were similar at baseline on measures of other key psychosocial factors that have been linked to cognitive changes, namely depression and fatigue[24], although sleep, another potential contributing factor, was not assessed.

Based on our findings and the two studies which have shown improved cognitive capacities with exercise in animals treated with chemotherapy[17, 18], the potential of aerobic exercise to improve cancer-associated cognitive symptoms remains worthy of further investigation. There are several lessons from this proof-of concept study to inform future human trials. First, at the time of recruitment there was limited evidence for an effective screening tool for persistent cancer-related cognitive changes. Researchers continue to test numerous meta-cognitive screening tools which could be useful in identifying patients in clinical practice and research studies such as the Patient's Assessment of Own Functioning Inventory (PAOFI), a self-report questionnaire that is positively associated with performance on objective neuropsychological testing in women who have been treatment for breast cancer[25]. This may be a potential screening tool for future trials. Second, , recruitment was a significant challenge for the trial. In addition to reporting persistent cognitive concerns following chemotherapy treatment for breast cancer, women had to be willing to be randomized potentially to the delayed exercise control group, be currently receiving anti-hormonal treatment and not be regularly physically active already. Future trials may want to consider a multiple site trial design and to focus on trial sites where participants are less likely to be physically active at baseline or re-assess the eligibility requirements, such as handling receipt of anti-hormonal treatment or not through stratification at enrollment rather than part of the eligibility critiera. Finally, in this study participants were eligible for inclusion up to three years following completion of adjuvant treatment. In considering design of future trials to examine the potential to address cognitive concern related to chemotherapy, it may be worth considering that there remains ongoing debate on the trajectory of improvement in cognitive symptoms following completion of treatment[39] and thus the most effective timing for intervention. One option for future trials may be to enroll women in an exercise intervention at the start of chemotherapy with the goal of limiting the development of cognitive issues or intervening just following completion of chemotherapy to stimulate or expedited recovery.

In conclusion, results of this proof-of-concept RCT support the potential of aerobic exercise to improve cognitive complaints in BCS, and can serve to inform the development of a future, definitive RCT. Further research into the sensitivity of outcome measures that capture cognitive complaints in BCS is also still needed.

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Table 1. Participant Characteristics

	All	**Control (n=9)**	**Exercise (n=10)**	***p****
	Mean (SD)	*Mean (SD)*	*Mean (SD)*	
Age (years)	52.4 (6.2)	51.4 (5.1)	53.2 (7.0)	0.54
Height (cm)	163.0 (6.3)	159.0 (8.0)	164.4 (7.3)	0.15
Weight (kg)	70.3 (15.4)	66.6 (16.7)	71.1 (15.9)	0.56
BMI (kg/m^2)	26.4 (5.5)	26.1 (5.5)	26.3 (5.7)	0.95
Usual physical activity (MET-hr/wk)	12.7 (9.8)	11.2 (7.0)	14.1 (12.0)	0.54
VO_2peak (ml/kg/min)	25.2 (6.5)	26.4 (5.7)	23.9 (7.0)	0.43
CES-D	15.5 (2.2)	15.4 (2.4)	15.6 (2.2)	0.88
STAS	21/7 (3.1)	21.6 (3.4)	21.8 (3.0)	0.87
FACT-F	109.9 (25.6)	113.0 (23.3)	107.1 (28.5)	0.63
Clinical Characteristics				
Time since completion of chemotherapy	11.2 (6.7)	11.1 (6.6)	11.5 (7.2)	0.90
	n (%)	*n (%)*	*n (%)*	*p*
Stage				0.06
I	0	0	0	
II	17 (90)	7(78)	10 (100)	
III	2 (10)	2(22)	0	
Chemotherapy Protocol				0.05
AC (all variants)	4 (22)	3 (38)	1 (10)	
DC	6 (33)	4 (50)	2 (20)	
FEC	8 (45)	1 (12)	7 (70)	
Radiation				0.25
Yes	17 (90)	7 (88)	10 (100)	
No	2 (10)	1 (12)	0	
Education Vocational training/ Some college or less	6 (32)	3 (33)	3 (30)	0.59
College Graduate	9 (47)	5 (55)	4 (40)	
Some Post graduate or more	4 (21)	1 (11)	3 (30)	
Employment Full-time	12 (63)	6 (67)	6 (60)	0.57
Part-time	1(5)	1 (11)	0	
Long-term disability	4 (21)	2 (22)	2 (20)	
Not employed or Retired	2 (11)	0	2 (20)	

**p*-value from independent samples t-test for continuous variables, and chi-square test for categorical variables

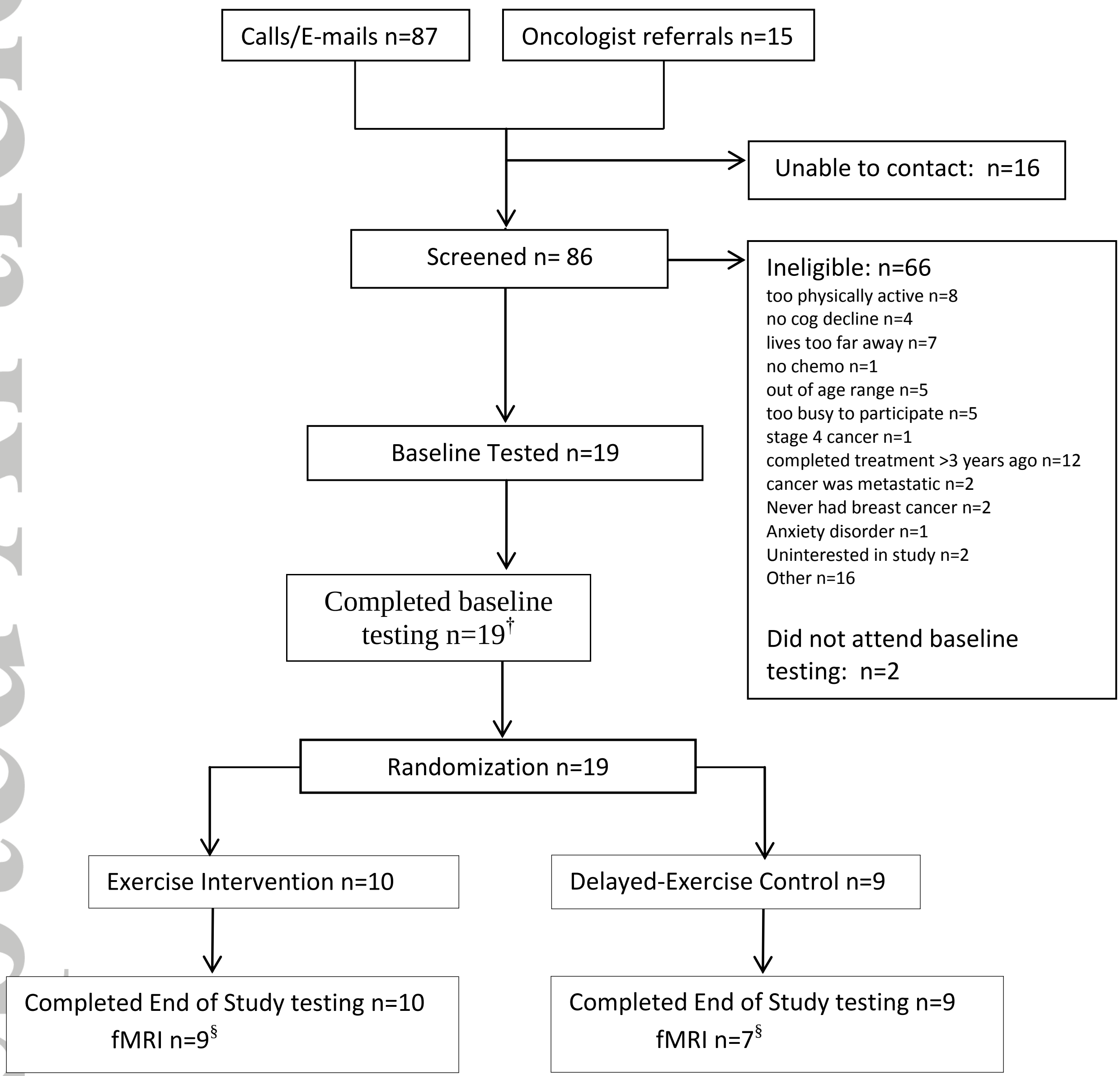
Legend: AC, Doxorubicin and Cyclophosphamide; DC, Docetaxel and Cyclophosphamide; FEC, 5-Flourouracil, Epirubicin, Cyclophosphamide.

Table 2. Changes in aerobic fitness, body weight, self-reported and objective cognitive outcomes by intervention group

	Control (CON)		Exercise (EX)		Adjusted difference between control and intervention group		Effect Size
	Baseline	**Δ (SD)**	**Baseline**	**Δ (SD)**	**Δ (95% CI)**	***p****	
	Mean (SD)		*Mean (SD)*				
VO_2 peak (ml/kg/min)	26.4 (5.7)	- 0.3 (1.8)	23.9 (7.0)	+ 3.4 (2.5)	+ 3.6 (1.4, 5.8)	**<0.01**	0.43
Weight (kg)	66.6 (16.7)	+ 0.2 (1.6)	71.1 (15.9)	- 0.2 (2.5)	- 0.4 (-2.6, 1.8)	0.70	0.10
Depression	15.4 (2.4)	- 0.3 (1.3)	15.6 (2.2)	- 1.3 (3.0)	- 0.9 (-3.9, 2.2)	0.56	0.02
Anxiety	21.6 (3.4)	- 0.7 (2.6)	21.8 (3.0)	- 0.3 (3.7)	+ 0.4 (-2.7, 3.6)	0.78	0.01
FACT-F TOI	73.6 (19.6)	+ 0.5 (11.4)	71.4 (21.1)	+ 4.7 (10.1)	+ 6.3 (-3.1, 15.8)	0.18	0.11
FACT- Cog							
Perceived Cog Impair.	28.8 (14.8)	+ 5.4 (9.5)	31.4 (13.8)	+ 8.8 (11.7)	+ 3.9 (-6.5, 14.4)	0.44	0.04
Impairment QoL	10.9 (2.4)	+ 0.6 (1.2)	7.9 (4.2)	+ 3.5 (3.5)	+ 1.8 (-1.0, 4.5)	0.19	0.11
Comments of Others	10.4 (5.3)	+ 1.0 (1.0)	11.7 (3.8)	+ 2.1 (2.3)	+ 1.5 (-0.3, 3.4)	0.10	0.18
Perceived Cog Abilities	12.0 (4.5)	+ 2.3 (4.0)	12.9 (3.8)	+ 3.5 (4.9)	+ 1.6 (-2.6, 5.7)	0.43	0.04
FAS	35.6 (8.0)	+ 3.6 (2.0)	37.1 (11.1)	+ 2.0 (5.2)	-1.5 (-7.0, 4.0)	0.56	0.02
Animals	18.0 (4.3)	- 2.4 (4.4)	12.6 (4.7)	+ 2.1 (3.0)	+ 3.0 (-1.2, 7.2)	0.15	0.13
Trails A (s)	33.8 (9.1)	+ 8.6 (10.4)	43.0 (12.4)	- 9.0 (10.1)	-14.2 (-24.6, -3.9)	**0.01**	0.35
Trails B (s)	77.9 (16.5)	+ 13.5 (41.8)	85.5 (31.1)	- 4.5 (29.5)	-18.6 (-55.0, 17.7)	0.29	0.07
Trails Diff. (s)	44.0 (12.6)	+ 4.8 (37.8)	42.5 (25.5)	+ 4.5 (32.2)	+0.1 (-34.7, 34.9)	0.99	0.00
HVLT							
Total recall	25.2 (4.0)	+ 1.0 (5.1)	23.5 (6.7)	+ 0.8 (7.3)	-1.5 (-6.3, 3.4)	0.52	0.03
Delayed recall	9.1 (2.6)	- 0.4 (3.2)	8.5 (3.1)	- 0.5 (3.0)	-0.5 (-2.7, 1.7)	0.62	0.02
Retention	92.3 (23.7)	- 7.9 (23.4)	98.5 (26.5)	- 15.2 (26.7)	-2.0 (-14.9, 11.0)	0.75	0.01
RDI	10.9 (0.9)	- 1.1 (1.3)	10.7 (1.2)	- 0.5 (1.8)	+0.5 (-0.8, 1.7)	0.47	0.03

Legend: *, p-value for difference in end of study values between groups after adjustment for baseline values using ANCOVA

Figure 1: Flow of participants through the trial



Legend: † *n=2 participants opted out of baseline fMRI;* §*n=3 participants opted out of end of study fMRI*